

Performance Metrics Assessment

Classification Model Evaluation

Confusion Matrix Components

- True Positives (TP) = 40
- True Negatives (TN) = 45
- False Positives (FP) = 5
- False Negatives (FN) = 10

Performance Metrics

Accuracy

$$\text{Accuracy} = \frac{40 + 45}{100} = 0.85 \quad (85\%)$$

Precision

$$\text{Precision} = \frac{40}{45} \approx 0.889$$

Recall

$$\text{Recall} = \frac{40}{50} = 0.80$$

F1 Score

$$\text{F1} \approx 0.842$$

Interpretation

- The model performs better on **non-spam detection** (90%) than spam detection (80%).
- **Precision** is more critical to avoid misclassifying important emails.
- Increasing **False Positives** lowers precision.
- Accuracy is unreliable for **imbalanced datasets**; F1-score is better.

Regression Model Evaluation

Error Table

Metrics

$$\text{MAE} = 16$$

$$\text{MSE} = 280$$

$$\text{RMSE} \approx 16.73$$

$$R^2 \text{ Score} = 0.944$$

House	Actual	Predicted	Error	Abs	Squared
1	200	180	20	20	400
2	150	160	-10	10	100
3	300	290	10	10	100
4	250	270	-20	20	400
5	100	120	-20	20	400

Interpretation

- **MSE/RMSE** penalize large errors more due to squaring.
- Low MAE but high RMSE = occasional large prediction errors.
- $R^2 \approx 1$ indicates a strong model fit.
- Negative R^2 means model is worse than predicting the mean.